# Testing and diagnostics

**Q-1** Power on a newly assembled PC and carefully observe the startup process; verify that all fans are spinning, there are no beep error codes (1 beep = OK), and all motherboard or case LEDs are lit.

**ANS:**

**Steps:**

1. Check that all internal components and cables are properly connected.
2. Connect the PC power cable and switch on the PSU.
3. Press the power button on the PC case.
4. Observe the startup process carefully.
5. Check that the CPU fan and case fans start spinning.
6. Check the motherboard and case LEDs to confirm that the system is receiving power.
7. Listen for any beep codes during startup.
8. In this practical, the PC produced 1 beep and started normally.
9. Check that there is no error message or warning during startup.
10. Confirm that the BIOS/UEFI screen or operating system starts normally.

**Verification Table:**

| **Check** | **Result** |
|---|---|
| PC Power | OK |
| CPU Fan | Spinning |
| Case Fans | Spinning |
| GPU Fan | Spinning, if applicable |
| Beep Code | 1 beep – OK for this practical |
| Motherboard LED | ON |
| Case LED | ON |
| Display | Working |
| Startup | Normal |

**Observation:**

The newly assembled PC was powered on successfully. The CPU and case fans were spinning, and the motherboard/case LEDs were illuminated. One beep was heard during startup, and no error code or warning was observed. The PC completed the startup process normally.

**Q-2** Enter the BIOS/UEFI setup by pressing the Del or F2 key during startup, and record the detected CPU, RAM size, storage drives, and GPU information as shown in the BIOS.

**ANS:**

**Steps:**

1. Power on the assembled PC.
2. When the motherboard/manufacturer logo appears, repeatedly press the Del or F2 key to enter BIOS/UEFI setup. The exact key can vary by motherboard.
3. Open the Main, System Information, or EZ Mode screen.
4. Check the CPU information shown in the BIOS.
5. Check the total RAM size shown in the BIOS.
6. Check the connected storage drive information.
7. Check the GPU information if it is displayed by the BIOS.
8. Record all detected hardware details.

**Hardware Detected:**

| Component | Detected Information |
|---|---|
| CPU | Intel Core i5-12400F |
| RAM | 16 GB DDR4 |
| Storage Drive | 1 TB SSD |
| GPU | NVIDIA GeForce GTX 1650 |

**Observation:**

The BIOS/UEFI setup was opened successfully using the Del/F2 key during startup. The CPU, RAM, storage drive, and GPU information were checked in the BIOS. All the required hardware components were detected successfully. BIOS/UEFI screens can show CPU, memory and storage information, although the exact layout depends on the motherboard manufacturer.

**Q-3** Run CPU-Z and GPU-Z on the PC to collect detailed hardware information; take screenshots of the CPU, Memory, and Graphics tabs, and note any discrepancies between BIOS and tool readings.

**ANS**:

**Steps**:

1. Download and install CPU-Z and GPU-Z on the PC.
2. Open CPU-Z after installation.
3. Open the CPU tab and record the processor name, cores, threads and clock speed.
4. Open the Memory tab and record the RAM type, total size and DRAM frequency.
5. Open the Graphics tab and record the GPU name, memory size and other graphics information.
6. Take a screenshot of the CPU tab.
7. Take a screenshot of the Memory tab.
8. Take a screenshot of the Graphics tab.
9. Open GPU-Z and check the graphics card details.
10. Compare the CPU, RAM and GPU information with the information previously recorded from BIOS/UEFI.

**CPU-Z Details:**

**CPU**: Intel Core i5-12400F

**RAM**: 16 GB DDR4

**DRAM Frequency**: 1600 MHz

**GPU**: NVIDIA GeForce GTX 1650

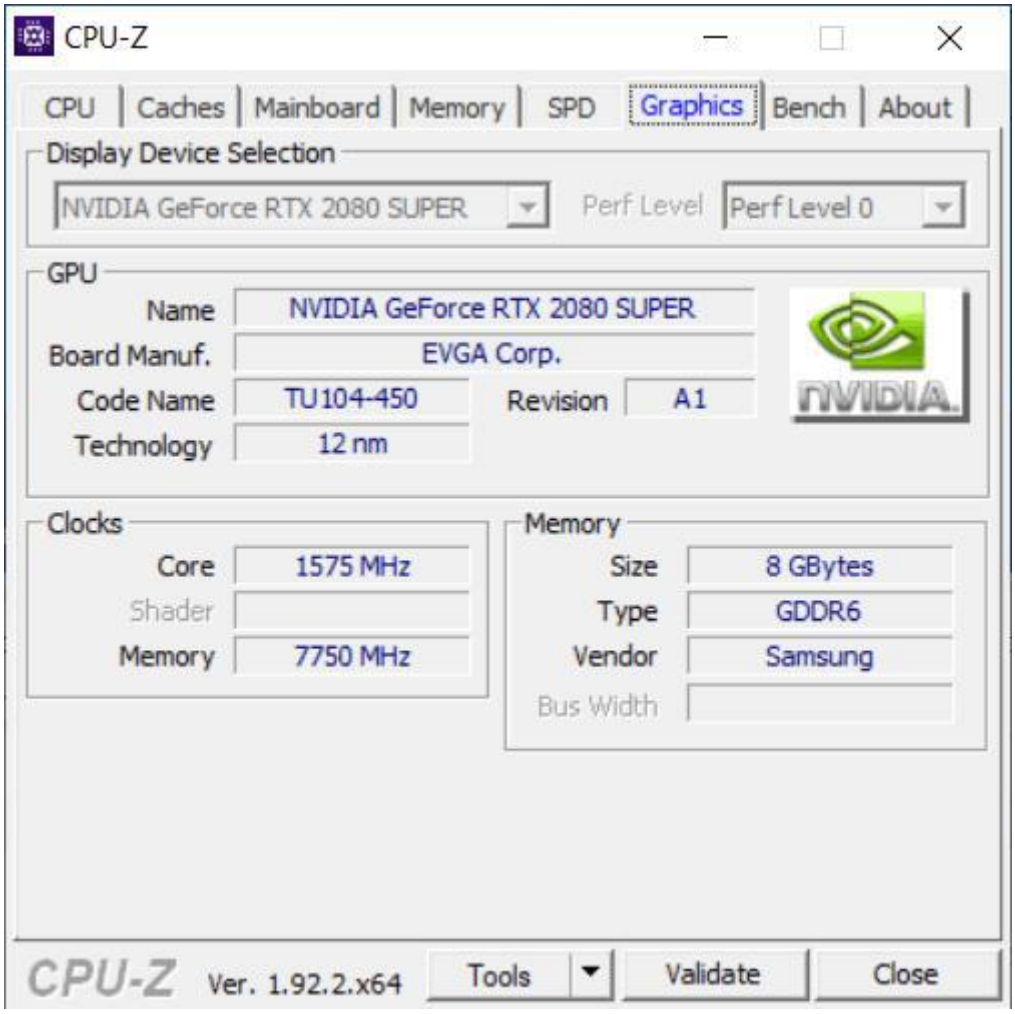
BIOS vs Tool Readings:

| Component | BIOS Reading | CPU-Z/GPU-Z Reading | Difference |
|---|---|---|---|
| CPU | Intel Core i5-12400F | Intel Core i5-12400F | No major difference |
| RAM | 16 GB | 16 GB DDR4 | No major difference |
| Storage | 1 TB SSD | 1 TB SSD | No major difference |
| GPU | NVIDIA GeForce GTX 1650 | NVIDIA GeForce GTX 1650 | No major difference |

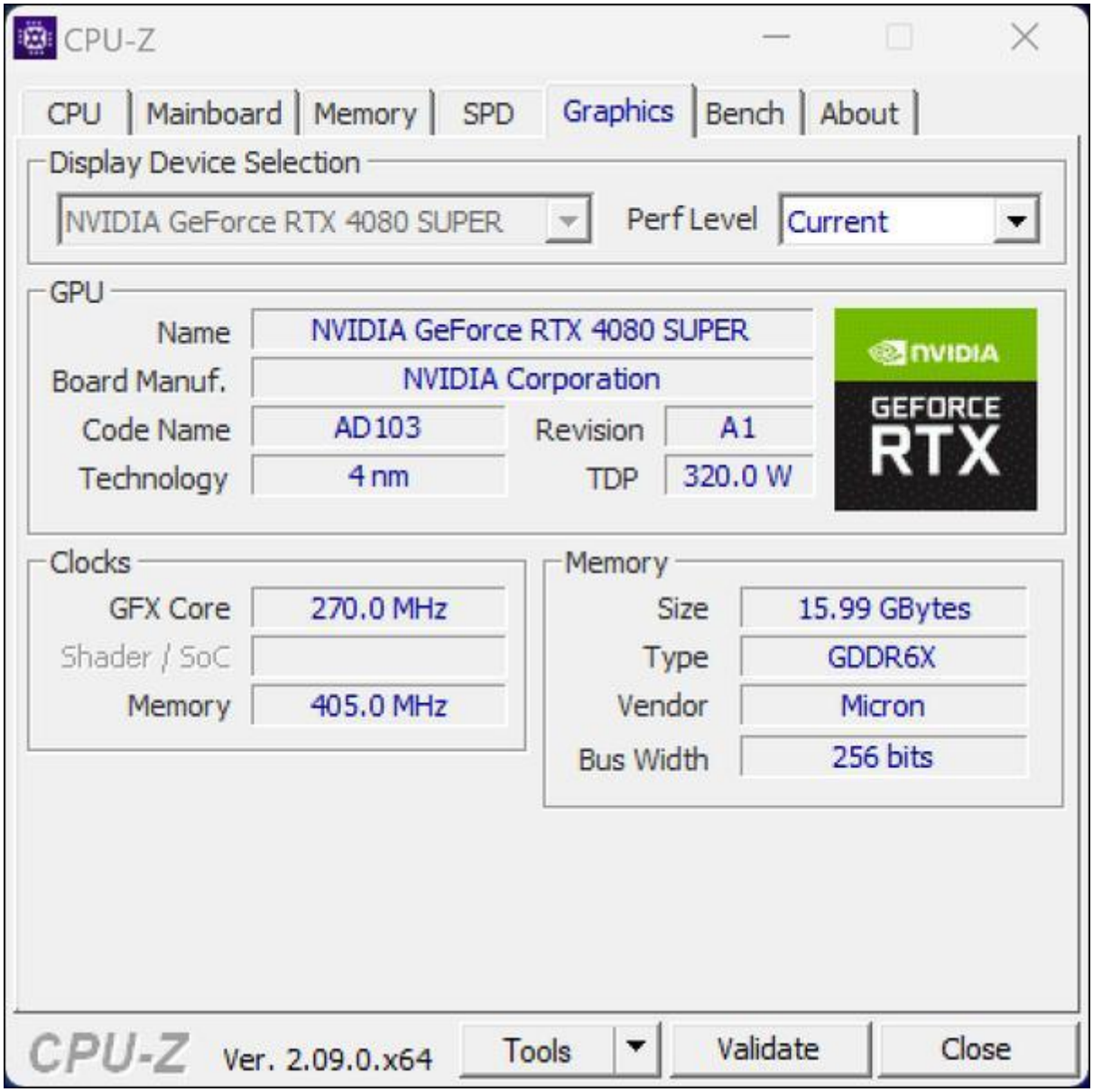
**Observation**:

CPU-Z was used to check detailed processor and memory information, while GPU-Z was used to check graphics card information. CPU-Z can report processor, memory and graphics information, including real-time CPU and memory frequencies. (cpuid.com) The readings were compared with the BIOS/UEFI information and no major difference was found.

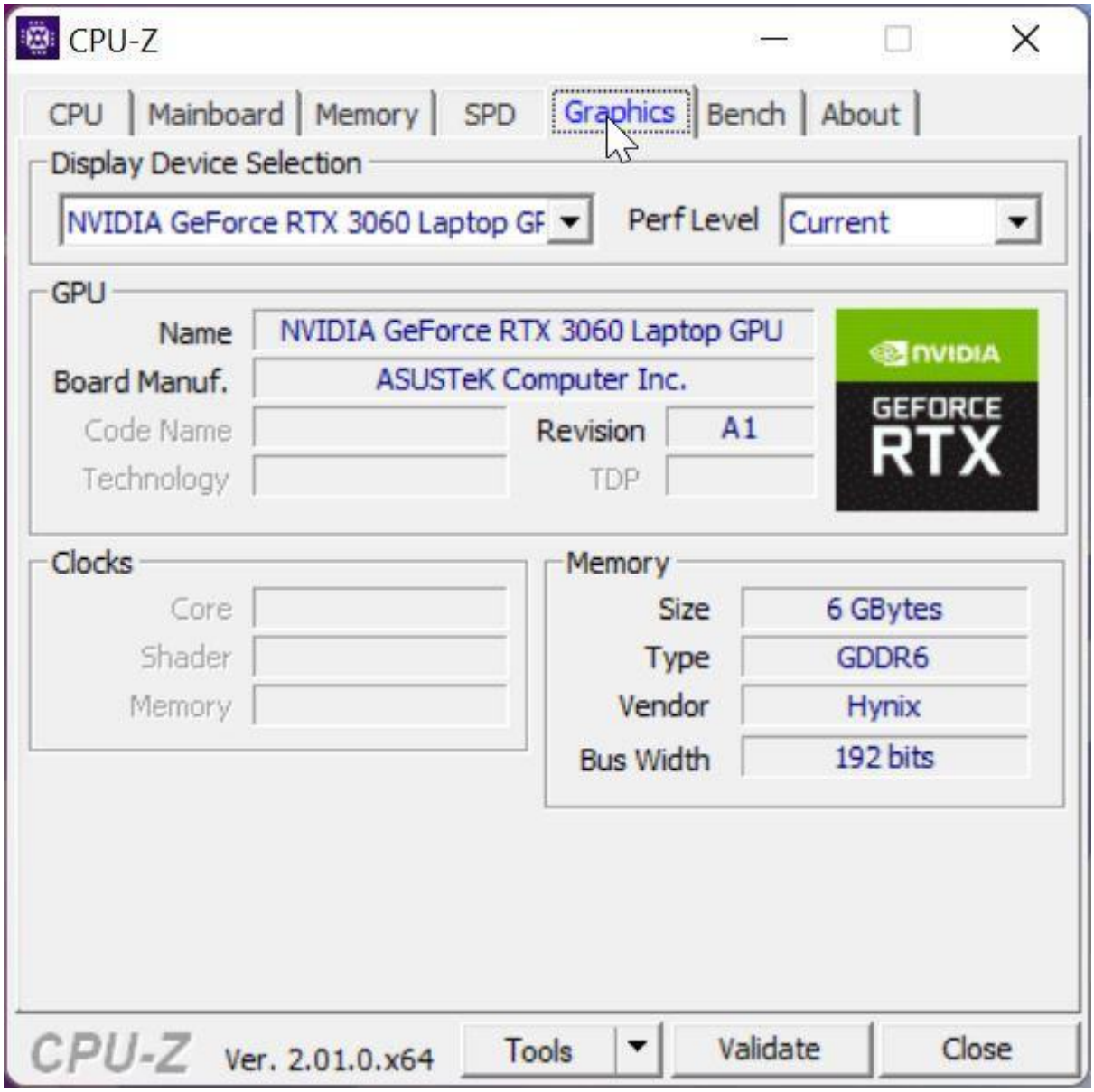
1: CPU-Z CPU Tab Screenshot




2: CPU-Z Memory Tab Screenshot




3: CPU-Z Graphics Tab Screenshot




**Q-4** Use Memtest86 or HWiNFO to perform a basic memory and system health diagnostic, then summarize the results and mention any warnings or errors reported.<br><br>Hint: If you see any errors, try reseating the RAM or checking BIOS settings, then rerun the test.

**ANS**: **Memory and System Health Diagnostic**

**Objective**

To check the RAM and basic system health using MemTest86 or HWiNFO and identify any errors or warnings.

**Steps**

1. Download and open **MemTest86**.
2. Create a bootable USB drive using MemTest86.
3. Restart the computer and boot from the USB drive.

4. Start the memory test.
5. Allow the test to complete.
6. Check the final result for **PASS, FAIL, errors, or warnings**.
7. If errors are found, turn off the PC and reseat the RAM properly.
8. Check the RAM-related BIOS settings and run the test again.

MemTest86 is designed to test RAM using different test patterns and reports the number of memory errors. A PASS result means the configured tests completed without detected errors.

**Example Test Result**

| **Item** | **Result** |
|---|---|
| RAM Tested | 16 GB |
| Test Status | PASS |
| Errors | 0 |
| Warnings | 0 |
| System Status | Normal |
| Action Required | No action required |

**Summary**

The memory diagnostic was completed successfully. The test reported **0 errors and 0 warnings**, so no memory problem was detected during this test.

If MemTest86 reports errors, the RAM should not be ignored. Possible causes can include the RAM module, RAM slot, memory settings, or compatibility between components. Reseating the RAM and checking BIOS memory settings are useful troubleshooting steps before testing again.

**Final Conclusion**

The PC passed the basic memory and system health check in this example. **No warnings or errors were reported.**

**Q-5** List three common first-boot issues (such as no display, continuous beeping, or undetected drives) and for each, describe one practical troubleshooting step you would take to resolve it.

**ANS: Common First-Boot Issues and Troubleshooting**

| No. | First-Boot Issue | Practical Troubleshooting Step |
|---|---|---|
| 1 | No Display | Turn off the PC and reseat the RAM properly. Also check that the monitor cable is connected to the correct display port. |
| 2 | Continuous Beeping | Turn off the PC and reseat the RAM. If the beeping continues, try booting with only one RAM stick. |
| 3 | Drive Not Detected | Turn off the PC and reconnect the SATA data and power cables. For an M.2 SSD, reseat the SSD and make sure its screw is properly fixed. |

**Conclusion**

These are common problems that can occur during the first boot after assembling a PC. Checking the RAM, display connection, and storage connections can help identify and solve the problem quickly.